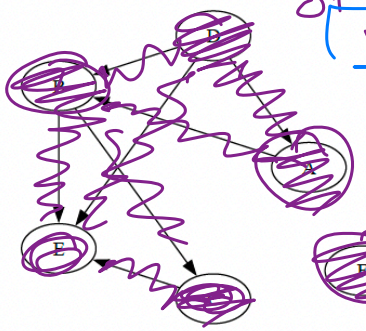
1. (3 points) Select all valid topological orders for the graph below.

A topological order is an order of all of the n nodes $v_1 v_2 \cdots v_n$ so that for every edge $v_i \to v_j$, $v_i < v_j$ in the order



- *ABCDEFG* ✗
- *GFEDCBA* ✗
- *DAFBECG* ✗
- *FDGABEC* ✗
- *FG* ✗ all nodes

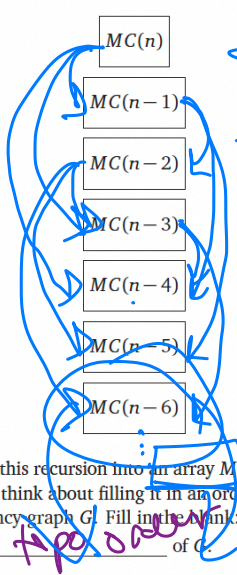
→ F D G, A B C E

2. (2 points) Which type of graph has the larger number of distinct topo orders?

- A directed graph with n nodes and zero edges — all orders are topo orders
- A directed tree with n nodes and $n-1$ edges
- A directed graph with n nodes and $n \cdot (n+1)/2$ edges (a complete garph) but no cycles — only one topo order

(a) Fill in all of the edges in this example (assume no nodes are base cases):

dependency graph is node for every subproblem, edge from s.p. u to s.p. v if u depends on v



MC(i) = max(MC(i−1)+1, MC(i−3)+1, MC(i−4)+1)

(b) Now consider memoizing this recursion into an array M. We know we can fill M from left to right, but now let's think about filling it in an order determined by some graph structure in the dependency graph G. Fill in the blank:
We fill M in a rev. topological order of G.

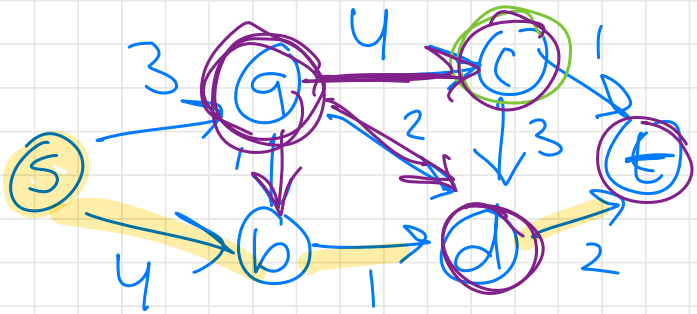
is a rev. topo order a topo order on a reversal of G?



Let H be the reversal of G.

a topo order of H is a reversal of a topo order of G.

Find the ~~longest~~ heaviest path from s to t in a DAG.



$LLP(a) = \max\{LLP(c)+4,\ LLP(d)+2,\ LLP(b)+1\}$

Greedy: start from s
choose heaviest edge
follow it to next node
from there, choose next heaviest edge
:
until reach t

Let $LLP(v)$ denote the length of the longest path from v to t.

want $LLP(s)$

$$LLP(v) = \begin{cases} 0 & \text{if } v = t \\ \max \text{ of } (LLP(u)\ \ \text{otherwise} \end{cases}$$

$+ \ell(u \to v)$
for all u so
that $v \Rightarrow u$ is
an edge)